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Braking a Car

Subject & topics: Physics | Mechanics | Dynamics **Stage & difficulty:** A Level C1

When a car of mass $1000\,\mathrm{kg}$ is travelling along a level road at a steady speed of $20\,\mathrm{m\,s^{-1}}$, its engine is working at $18\,\mathrm{kW}$.

Part A

Resistive force

Find the magnitude of the resistive force due to friction, which may be taken to be constant.

Part B

Braking force on the level

The engine is suddenly disconnected and the brakes applied, and the car comes to rest in $50\,\mathrm{m}$. Find the force, assumed constant, exerted by the brakes.

Part C

Braking force on a slope

Find also the distance in which the car, travelling at $20\,\mathrm{m\,s^{-1}}$, would come to rest if the engine were disconnected and the same braking force applied on an upward incline of angle , where $\sin \; = \frac{1}{20}$.

Part D

Braking force downhill

By how much does this braking distance change if the car is travelling down the same hill at $20\,\mathrm{m\,s^{-1}}$?

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Poisson Distribution - Woodland

Subject & topics: Maths | Statistics | Probability **Stage & difficulty:** Further A C1, University P1

There are on average 140 trees per hectare ($10^4\,\mathrm{m}^2$) in a wood, 30% of which are oak trees. On the assumption that the distribution of trees in the wood follows a Poisson distribution, answer the following.

Part A

Less than 4 oaks

Find the probability that there are less than 4 oaks in an area of $1000\,\mathrm{m}^2$. Give your answer to 4 s.f.

Part B

Most probable number of other species

Find the most probable number of trees which are not oaks in an area of $1000\,\mathrm{m}^2$.

Part C

One oak in each of 3 areas

Three $500\,\mathrm{m}^2$ areas are selected at random. Find the probability that there will be exactly 1 oak in each of them. Give your answer to 3 s.f.

Part D

Number of areas with exactly 6 oaks

It is known that in the wood 30 areas of $1000\,\mathrm{m}^2$ contain 6 or more oaks. Obtain an estimate of the number of these which contain exactly 6 oaks, giving your answer to the nearest integer.

Part E

4 selected areas

Four areas of $500\,\mathrm{m}^2$ are selected at random. Find the overall probability that two of the areas contain exactly 8 trees, one contains more than 8 trees and one contains fewer than 8 trees. Give your answer to 3 s.f.

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Hypothesis Testing: Exams

Normal Distribution

Subject & topics: Maths | Statistics | Hypothesis Tests **Stage & difficulty:** Further A P2

An examination board is developing a new syllabus and wants to know if the question papers are the right length. A random sample of 50 candidates was given a pre-test on a dummy paper. The times, $t\,\mathrm{minutes}$, taken by these candidates to complete the paper can be summarised by

$$n = 50, \quad \Sigma t = 4050, \quad \Sigma t^2 = 329\,800.$$

Assume that times are normally distributed.

Part A

Not completed within $90\,\mathrm{minutes}$

Estimate the proportion of candidates that could not complete the paper within $90\,\mathrm{minutes}$. Give your answer to 3 s.f.

Part B

Hypothesis test

Test, at the 10% significance level, whether the mean time for all candidates to complete this paper is $80\,\mathrm{minutes}$. Use a two-tailed test. Fill in the gaps below.

Let μ be the mean time to complete the test. The null and alternative hypotheses are:

$$\mathrm{H}_0 : \mu = \boxed{} \qquad \mathrm{H}_1 : \mu \boxed{}$$

Assuming that the null hypothesis is true, the distribution of sample means $\overline{T} \sim \mathrm{N}(\boxed{}, \boxed{})$.

The test statistic is $\mathrm{P}(\overline{T} \geq \boxed{}) = \boxed{}$.

For a two-tailed test at the 10% significance level, we find that the test statistic is $\boxed{}$ $\boxed{}$.

Therefore, $\boxed{}$ the null hypothesis. There $\boxed{}$ evidence to suggest that the mean time for all candidates to take this paper is not equal to $80\,\mathrm{minutes}$.

Items:

0.118 | 35 | > 80 | 0.250 | greater than | less than | 81 | reject | $\frac{250}{7}$ | 0.05 | < 80 | 0.0118 | $\neq 80$ | 80 | 0.7 | is | $\frac{5}{7}$ | 0.1 | is insufficient | do not reject | 0.116

Part C

Assumptions

Explain whether the assumption that times are normally distributed is necessary in answering each part. Fill in the gaps below.

In part A, the assumption that the times are normally distributed ______ necessary. An unbiased estimate of the population mean and the population variance can be made from the sample ______ this assumption. However, calculating a ______ from these values requires knowledge of the distribution.

In part B, the assumption that the times are normally distributed ______ necessary. The central limit theorem states that the sample means for samples taken from ______ distribution will be approximately normally distributed as long as the sample size is sufficiently ______, which is the case in this question ($n = 50$).

Items:

with | without | small | is not | probability | a normal | a discrete | large | any | is | a continuous

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Restitution: Sphere Collision 2

Subject & topics: Physics | Mechanics | Dynamics **Stage & difficulty:** Further A C2

Two uniform smooth spheres A and B of equal radius are moving on a horizontal surface when they collide. A has mass $0.10\,\mathrm{kg}$ and B has mass $0.40\,\mathrm{kg}$. Immediately before the collision A is moving with speed $2.8\,\mathrm{m\,s^{-1}}$ along the line of centres, and B is moving with speed $1.0\,\mathrm{m\,s^{-1}}$ at an angle θ to the line of centres, where $\cos\theta = 0.80$, as shown in **Figure 1**. Immediately after the collision A is stationary.

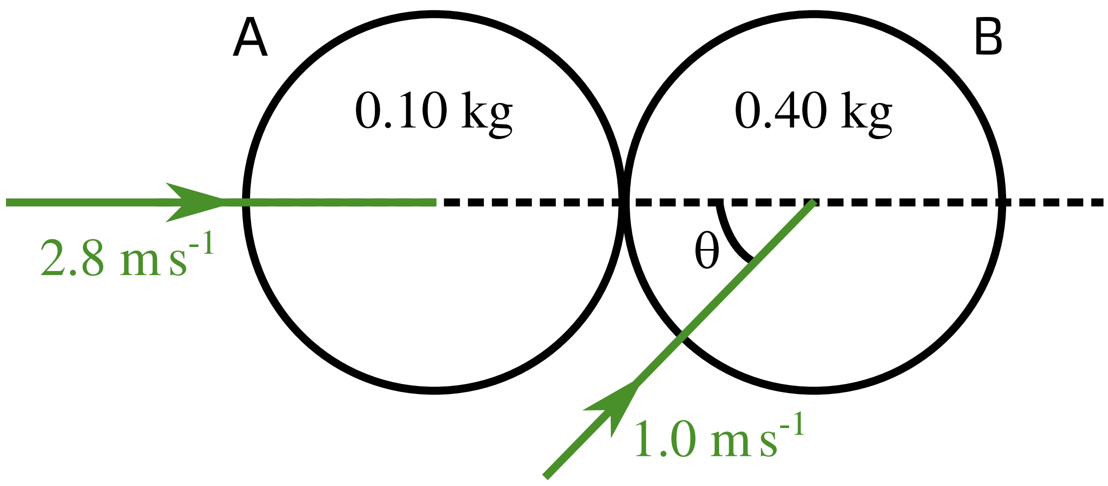

Figure 1: The spheres A and B as they collide.

Part A
Coefficient of restitution

Find the value of the coefficient of restitution between A and B.

Part B
Change in direction

Find the angle turned through by the direction of motion of B as a result of the collision.

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Continuous Random Variables 7

Subject & topics: Maths | Statistics | Random Variables **Stage & difficulty:** Further A P2

The probability density function $f(x)$ for a random variable X is given by

$$f(x) = \begin{cases} \alpha & 0 \leq x \leq \frac{b}{2}, \\ 2\alpha & \frac{b}{2} < x \leq b, \\ 0 & \text{otherwise,} \end{cases}$$

where α and b are constants.

Part A

Find α

Find an expression for α in terms of b.

The following symbols may be useful: `alpha`, `b`

Part B

Find $\mathrm{E}(X)$

Find the expectation of X in terms of b.

The following symbols may be useful: `b`

Part C

Find $\mathrm{E}(X^2)$

Find the expectation of X^2 in terms of b.

The following symbols may be useful: b

Part D

Standard deviation

Find the standard deviation of X in terms of b.

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Two Spinning Balls

Subject & topics: Physics | Mechanics | Circular Motion **Stage & difficulty:** A Level C2

A ball of mass m_1 is suspended from a fixed point O by a light string of length $l = 5.0\,\mathrm{m}$ and a second ball of mass m_2 is suspended from the first by a light string of the same length.

When the system is rotating steadily with angular velocity ω about the vertical through O , the balls describe horizontal circles of radii $3.0\,\mathrm{m}$ and $7.0\,\mathrm{m}$ with their centres $4.0\,\mathrm{m}$ and $7.0\,\mathrm{m}$ respectively below O .

Part A

Angular speed

Find the value of ω

Part B

Ratio of masses

What is the ratio of the masses of the balls? Express your answer as a decimal greater than one.

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isaac You work it out.

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Chi-squared Tests: Pension

Subject & topics: Maths | Statistics | Hypothesis Tests **Stage & difficulty:** Further A P2

A survey is conducted of customers using a Post Office. Starting when the Post Office opens in the morning, a count is made of the number of customers up to and including the first person to collect their old age pension. This is repeated on each of a total of 40 different days and the results are summarised below.

Number of customers	1	2	3	4	5	6	7	8	9	10	≥ 11
Frequency	23	5	3	3	2	1	0	1	0	2	0

It is thought that this distribution may be modelled by a geometric distribution with parameter p, where p is the probability that a customer is collecting their old age pension.

Part A
Mean

Calculate the mean using the data above.

Part B
Value of p

Calculate the value of p from the data above, giving your answer as a fraction.

The following symbols may be useful: p

Part C

χ^2 statistic

It is intended to test, at the 5% significance level, the goodness of fit of the model to the data.

Using the proposed geometric model, calculate the χ^2 statistic for these data. Give your answer to 4 s.f.

Part D

Hypothesis test

Carry out the test at the 5% significance level. Fill in the gaps below.

The null and alternative hypotheses are

H_0 : The data [ ] the proposed model.

H_1 : The data [ ] the proposed model.

The appropriate critical value $\chi^2_{\text{crit}} =$ [ ].

We see that χ^2 [ ] χ^2_{crit}. Therefore we [ ] the null hypothesis. There [ ] evidence to suggest that the data do not fit the proposed model.

Items:

$>$ | 6.251 | is insufficient | do not fit | 9.488 | $=$ | 5.991 | reject | $<$ | is | fit | 4.605 | do not reject | 7.378 | 9.348 | 7.815

Adapted with permission from UCLES, A Level, January 2003 S3, Paper 2643, Question 5

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isaac You work it out.

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Centre of Mass: Lamina 2

Subject & topics: Physics | Mechanics | Statics **Stage & difficulty:** Further A P2

Figure 1 shows a uniform lamina BCD in the shape of a quarter circle of radius $6.0\,\mathrm{cm}$.

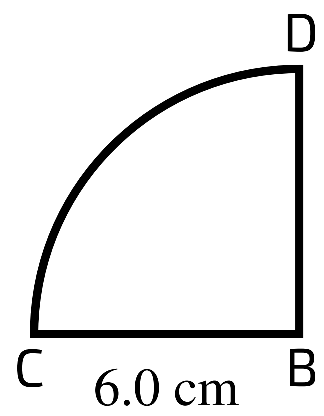

Figure 1: Lamina BCD.

Part A

Centre of mass of BCD

Find the distance of the centre of mass of the lamina from B.

Part B

Distance from BD

A uniform rectangular lamina ABDE is such that AB is $12\,\mathrm{cm}$ and AE is $6.0\,\mathrm{cm}$. A single plane object ABCDE is formed by attaching the rectangular lamina ABDE to the lamina BCD along BD, as shown in **Figure 2**. The mass of ABDE is $3.0\,\mathrm{kg}$ and the mass of BCD is $2.0\,\mathrm{kg}$.

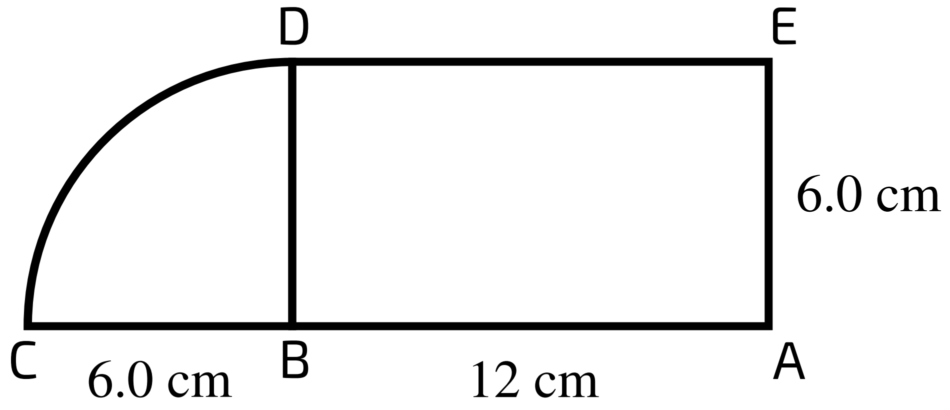

Figure 2: Object ABCDE, formed by joining laminas BCD and ABDE.

Find the distance of the centre of mass of the object ABCDE from BD.

Part C

Distance from AC

Find the distance of the centre of mass of the object ABCDE from AC.

Part D

Angle to vertical

The object ABCDE is freely suspended at C and rests in equilibrium.

Calculate the angle that AC makes with the vertical.

Adapted with permission from UCLES A Level, June 2009, Paper 4729, Question 5.

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Differential Equations: General Applications 2ii

Subject & topics: Maths **Stage & difficulty:** Further A P2

A particle P of mass $0.2\,\mathrm{kg}$ is suspended from a fixed point O by a light elastic string of natural length $0.7\,\mathrm{m}$ and modulus of elasticity $3.5\,\mathrm{N}$. P is at the equilibrium position when it is projected vertically downwards with speed $1.6\,\mathrm{m\,s^{-1}}$. At time $t\,\mathrm{s}$ after being set in motion P is $x\,\mathrm{m}$ below the equilibrium position and has velocity $v\,\mathrm{m\,s^{-1}}$.

The tension, T, in the string is expressed as

$$T = \frac{3.5(0.392 + x)}{0.7}\,\mathrm{N}$$

The equilibrium position of P is $1.092\,\mathrm{m}$ below O, and the strength of gravity is $9.8\,\mathrm{N\,kg^{-1}}$.

Part A

SHM

Prove that P moves with simple harmonic motion, and calculate the amplitude.

Part B

Find x

Calculate x when $t = 0.4$. Give your answer to 3 significant figures.

Part C

Find v

Calculate the velocity of P when $t = 0.4$. Give your answer to 3 significant figures.

Adapted with permission from UCLES, A Level, January 2007 , Paper 4730, Question 4.

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t-tests: Oak Leaves

Subject & topics: Maths | Statistics | Hypothesis Tests **Stage & difficulty:** Further A P3

The leaves from oak trees growing in two different areas A and B are being measured. The lengths, in cm, of a random sample of 7 oak leaves from area A are:

$$6.2, \quad 8.3, \quad 7.8, \quad 9.3, \quad 10.2, \quad 8.4, \quad 7.2$$

Part A

Confidence interval

Assuming that the distribution is normal, find a 95% confidence interval for the mean length of oak leaves from area A.

(,)

Part B

Assumptions

The lengths, in cm, of a random sample of 5 oak leaves from area B are:

$$5.9, \quad 7.4, \quad 6.8, \quad 8.2, \quad 8.7$$

State two assumptions needed to carry out a suitable t-test on the difference between the mean lengths of oak leaves from areas A and B.

- [ ] The length of oak leaves from area B follow a uniform distribution.
- [ ] The lengths of oak leaves from areas A and B have equal variances.
- [ ] The length of oak leaves from area B follow a normal distribution.
- [ ] The lengths of oak leaves from areas A and B have equal means.
- [ ] The length of oak leaves from area B follow a geometric distribution.

Part C

Hypothesis test

Test, at the 5% significance level, whether the mean length of oak leaves from area A is greater than the mean length of oak leaves from area B.

The null and alternative hypotheses are:

$$\mathrm{H}_0 : \mu_a \square \mu_b \qquad \mathrm{H}_1 : \mu_a \square \mu_b$$

Calculating the difference as $\overline{a} - \overline{b}$, the test statistic, $t =$ ___. The critical value is $t_{\text{crit}} =$ ___.

Comparing these, we find that t ___ t_{crit}.

Therefore we ___ H_0 at the 5% level. There ___ evidence to suggest that the mean length of oak leaves from area A is greater than the mean length of oak leaves from area B.

Items:

reject | 0.889 | is insufficient | $<$ | 1.812 | $=$ | do not reject | is | 2.179 | 2.228 | 1.782 | 1.102 | 1.208 | 1.137 | 1.140 | $\neq$ | $>$

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